

PACKAGE INSERT

1. NAME OF THE MEDICINAL PRODUCT

SUTYKA 12.5 (Sunitinib Hard Capsules 12.5 mg)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

12.5 mg hard capsules

Each capsule contains sunitinib malate, equivalent to 12.5 mg of sunitinib. For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Hard capsule.

Sunitinib 12.5 mg hard capsules

Yellow granular powder filled in Size "4" hard gelatin capsules with orange opaque body imprinted with "6" and orange opaque cap imprinted with "MS" in white ink, free from physical defects.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Gastrointestinal Stromal Tumour (GIST)

Sunitinib is indicated for the treatment of unresectable and/or metastatic malignant gastrointestinal stromal tumour (GIST) in adults after failure of imatinib treatment due to resistance or intolerance.

Metastatic Renal Cell Carcinoma (MRCC)

Sunitinib is indicated for the treatment of advanced/metastatic renal cell carcinoma (MRCC) in adults.

Pancreatic Neuroendocrine Tumours (pNET)

Sunitinib is indicated for the treatment of unresectable or metastatic, well-differentiated pancreatic neuroendocrine tumours (pNET) with disease progression in adults.

4.2 Posology and method of administration

Therapy with sunitinib should be initiated by a physician experienced in the administration of anticancer agents.

Posology

For GIST and MRCC, the recommended dose of sunitinib is 50 mg taken orally once daily, for 4 consecutive weeks, followed by a 2-week rest period (Schedule 4/2) to comprise a complete cycle of 6 weeks.

For pNET, the recommended dose of sunitinib is 37.5 mg taken orally once daily without a scheduled rest period.

Dose adjustments

Safety and tolerability

For GIST and MRCC, dose modifications in 12.5 mg steps may be applied based on individual safety and tolerability. Daily dose should not exceed 75 mg nor be decreased below 25 mg.

For pNET, dose modification in 12.5 mg steps may be applied based on individual safety and tolerability. The maximum dose administered in the Phase 3 pNET study was 50 mg daily.

Dose interruptions may be required based on individual safety and tolerability.

CYP3A4 inhibitors/inducers

Co-administration of sunitinib with potent CYP3A4 inducers, such as rifampicin, should be avoided (see sections 4.4 and 4.5). If this is not possible, the dose of sunitinib may need to be increased in 12.5 mg steps (up to 87.5 mg per day for GIST and MRCC or 62.5 mg per day for pNET) based on careful monitoring of tolerability.

Co-administration of sunitinib with potent CYP3A4 inhibitors, such as ketoconazole, should be avoided (see sections 4.4 and 4.5). If this is not possible, the dose of sunitinib may need to be reduced to a minimum of 37.5 mg daily for GIST and MRCC or 25 mg daily for pNET, based on careful monitoring of tolerability.

Selection of an alternative concomitant medicinal product with no or minimal potential to induce or inhibit CYP3A4 should be considered.

Special populations

Paediatric population

The safety and efficacy of sunitinib in patients below 18 years of age have not been established.

Currently available data are described in sections 4.8, 5.1, and 5.2 but no recommendation on a posology can be made.

Elderly

Approximately one-third of the patients in clinical studies who received sunitinib were 65 years of age or over. No significant differences in safety or efficacy were observed between younger and older patients.

Hepatic impairment

No starting dose adjustment is recommended when administering sunitinib to patients with mild or moderate (Child-Pugh class A and B) hepatic impairment. Sunitinib has not been studied in subjects with severe (Child-Pugh class C) hepatic impairment and therefore its use in patients with severe hepatic impairment cannot be recommended (see section 5.2).

Renal impairment

No starting dose adjustment is required when administering sunitinib to patients with renal impairment (mild-severe) or with end-stage renal disease (ESRD) on haemodialysis. Subsequent dose adjustments should be based on individual safety and tolerability (see section 5.2).

Method of administration

Sunitinib is for oral administration. It may be taken with or without food.

If a dose is missed, the patient should not be given an additional dose. The patient should take the usual prescribed dose on the following day.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

4.4 Special warnings and precautions for use

Co-administration with potent CYP3A4 inducers should be avoided because it may decrease sunitinib plasma concentration (see sections 4.2 and 4.5).

Co-administration with potent CYP3A4 inhibitors should be avoided because it may increase the plasma concentration of sunitinib (see sections 4.2 and 4.5).

Skin and tissue disorders

Patients should be advised that depigmentation of the hair or skin may occur during treatment with sunitinib. Other possible dermatological effects may include dryness, thickness or cracking of the skin, blisters, or rash on the palms of the hands and soles of the feet.

The above reactions were not cumulative, were typically reversible, and generally did not result in treatment discontinuation. Cases of pyoderma gangrenosum, generally reversible after discontinuation of sunitinib, have been reported. Severe cutaneous reactions have been reported, including cases of erythema multiforme (EM), cases suggestive of Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN), some of which were fatal. If signs or symptoms of SJS, TEN, or EM (e.g., progressive skin rash often with blisters or mucosal lesions) are present, sunitinib treatment should be discontinued. If the diagnosis of SJS or TEN is confirmed, treatment must not be restarted. In some cases of suspected EM, patients tolerated the reintroduction of sunitinib therapy at a lower dose after resolution of the reaction; some of these patients also received concomitant treatment with corticosteroids or antihistamines (see section 4.8).

Haemorrhage and tumour bleeding

Haemorrhagic events, some of which were fatal, reported in clinical studies with sunitinib and during postmarketing surveillance have included gastrointestinal, respiratory, urinary tract, and brain haemorrhages (see section 4.8).

Routine assessment of bleeding events should include complete blood counts and physical examination.

Epistaxis was the most common haemorrhagic adverse reaction, having been reported for approximately half of the patients with solid tumours who experienced haemorrhagic events. Some of the epistaxis events were severe, but very rarely fatal.

Events of tumour haemorrhage, sometimes associated with tumour necrosis, have been reported; some of these haemorrhagic events were fatal.

Tumour haemorrhage may occur suddenly, and in the case of pulmonary tumours, may present as severe and life-threatening haemoptysis or pulmonary haemorrhage. Cases of pulmonary haemorrhage, some with a fatal outcome, have been observed in clinical trials and have been reported in postmarketing experience in patients treated with sunitinib for MRCC, GIST, and lung cancer. Sunitinib is not approved for use in patients with lung cancer.

Patients receiving concomitant treatment with anticoagulants (e.g., warfarin, acenocoumarole) may be periodically monitored by complete blood counts (platelets), coagulation factors (PT/INR), and physical examination.

Gastrointestinal disorders

Diarrhoea, nausea/vomiting, abdominal pain, dyspepsia, and stomatitis/oral pain were the most commonly reported gastrointestinal adverse reactions; oesophagitis events have been also reported (see section 4.8).

Supportive care for gastrointestinal adverse reactions requiring treatment may include medicinal products with antiemetic, antidiarrhoeal, or antacid properties.

Serious, sometimes fatal gastrointestinal complications including gastrointestinal perforation were reported in patients with intra-abdominal malignancies treated with sunitinib.

Hypertension

Hypertension has been reported in association with sunitinib, including severe hypertension (> 200 mmHg systolic or 110 mmHg diastolic). Patients should be screened for hypertension and controlled as appropriate. Temporary suspension is recommended in patients with severe hypertension that is not controlled with medical management. Treatment may be resumed once hypertension is appropriately controlled (see section 4.8).

Haematological disorders

Decreased absolute neutrophil counts and decreased platelet counts were reported in association with sunitinib (see section 4.8). The above events were not cumulative, were typically reversible, and generally did not result in treatment discontinuation. None of these events in the Phase 3 studies were fatal, but rare fatal haematological events, including haemorrhage associated with thrombocytopenia and neutropenic infections, have been reported during postmarketing surveillance.

Anaemia has been observed to occur early as well as late during treatment with sunitinib.

Complete blood counts should be performed at the beginning of each treatment cycle for patients receiving treatment with sunitinib (see section 4.8).

Cardiac disorders

Cardiovascular events, including heart failure, cardiomyopathy, left ventricular ejection fraction decline to below the lower limit of normal, myocarditis, myocardial ischaemia and myocardial infarction, some of which were fatal, have been reported in patients treated with sunitinib. These data suggest that sunitinib increases the risk of cardiomyopathy. No specific additional risk factors for sunitinib-induced cardiomyopathy apart from the drug-specific effect have been identified in the treated patients. Use sunitinib with caution in patients who are at risk for, or who have a history of, these events (see section 4.8).

Patients who presented with cardiac events within 12 months prior to sunitinib administration, such as myocardial infarction (including severe/unstable angina), coronary/peripheral artery bypass graft, symptomatic congestive heart failure (CHF), cerebrovascular accident or transient ischaemic attack, or pulmonary embolism were excluded from all sunitinib clinical studies. It is unknown whether patients with these concomitant conditions may be at a higher risk of developing sunitinib-related left ventricular dysfunction.

Physicians are advised to weigh this risk against the potential benefits of sunitinib. Patients should be carefully monitored for clinical signs and symptoms of CHF while receiving sunitinib especially patients with cardiac risk factors and/or history of coronary artery disease. Baseline and periodic evaluations of LVEF should also be considered while the patient is receiving sunitinib. In patients without cardiac risk factors, a baseline evaluation of ejection fraction should be considered.

In the presence of clinical manifestations of CHF, discontinuation of sunitinib is recommended. The administration of sunitinib should be interrupted and/or the dose reduced in patients without clinical evidence of CHF but with an ejection fraction < 50% and > 20% below baseline.

QT interval prolongation

Prolongation of QT interval and Torsade de pointes have been observed in sunitinib-exposed patients. QT interval prolongation may lead to an increased risk of ventricular arrhythmias including Torsade de pointes.

Sunitinib should be used with caution in patients with a known history of QT interval prolongation, patients who are taking antiarrhythmics or medicinal products that can prolong QT interval, or patients with relevant pre-existing cardiac disease, bradycardia, or electrolyte disturbances. Concomitant administration of sunitinib with potent CYP3A4 inhibitors should be limited because of the possible increase in sunitinib plasma concentrations (see sections 4.2, 4.5 and 4.8).

Venous thromboembolic events

Treatment-related venous thromboembolic events were reported in patients who received sunitinib including deep venous thrombosis and pulmonary embolism (see section 4.8). Cases of pulmonary embolism with fatal outcome have been observed in postmarketing surveillance.

Arterial thromboembolic events

Cases of arterial thromboembolic events (ATE), sometimes fatal, have been reported in patients treated with sunitinib. The most frequent events included cerebrovascular accident, transient ischaemic attack, and cerebral infarction. Risk factors associated with ATE, in addition to the underlying malignant disease and age ≥ 65 years, included hypertension, diabetes mellitus, and prior thromboembolic disease.

Aneurysms and artery dissections

The use of vascular endothelial growth factor (VEGF) pathway inhibitors in patients with or without hypertension may promote the formation of aneurysms and/or artery dissections. Before initiating sunitinib, this risk should be carefully considered in patients with risk factors such as hypertension or history of aneurysm.

Thrombotic microangiopathy (TMA)

The diagnosis of TMA, including thrombotic thrombocytopenic purpura (TTP) and haemolytic uraemic syndrome (HUS), sometimes leading to renal failure or a fatal outcome, should be considered in the occurrence of haemolytic anaemia, thrombocytopenia, fatigue, fluctuating neurological manifestation, renal impairment, and fever. Sunitinib therapy should be discontinued in patients who develop TMA and prompt treatment is required. Reversal of the effects of TMA has been observed after treatment discontinuation (see section 4.8).

Thyroid dysfunction

Baseline laboratory measurement of thyroid function is recommended in all patients. Patients with pre-existing hypothyroidism or hyperthyroidism should be treated as per standard medical practice prior to the start of sunitinib treatment. During sunitinib treatment, routine monitoring of thyroid function should be performed every 3 months. In addition, patients should be observed closely for signs and symptoms of thyroid dysfunction during treatment, and patients who develop any signs and/or symptoms suggestive of thyroid dysfunction should have laboratory testing of thyroid function performed as clinically indicated. Patients who develop thyroid dysfunction should be treated as per standard medical practice.

Hypothyroidism has been observed to occur early as well as late during treatment with sunitinib (see section 4.8).

Pancreatitis

Increases in serum lipase and amylase activities were observed in patients with various solid tumours who received sunitinib. Increases in lipase activities were transient and were generally not accompanied by signs or symptoms of pancreatitis in subjects with various solid tumours (see section 4.8).

Cases of serious pancreatic events, some with fatal outcome, have been reported. If symptoms of pancreatitis are present, patients should have sunitinib discontinued and be provided with appropriate supportive care.

Hepatotoxicity

Hepatotoxicity has been observed in patients treated with sunitinib. Cases of hepatic failure, some with a fatal outcome, were observed in < 1% of solid tumour patients treated with sunitinib. Monitor liver function tests (alanine transaminase [ALT], aspartate transaminase [AST], bilirubin levels) before initiation of treatment, during each cycle of treatment, and as clinically indicated. If signs or symptoms of hepatic failure are present, sunitinib should be discontinued and appropriate supportive care should be provided (see section 4.8).

Renal function

Cases of renal impairment, renal failure and/or acute renal failure, in some cases with fatal outcome, have been reported (see section 4.8).

Risk factors associated with renal impairment/failure in patients receiving sunitinib included, in addition to underlying RCC, older age, diabetes mellitus, underlying renal impairment, cardiac failure, hypertension, sepsis, dehydration/hypovolaemia, and rhabdomyolysis.

The safety of continued sunitinib treatment in patients with moderate to severe proteinuria has

not been systematically evaluated.

Cases of proteinuria and rare cases of nephrotic syndrome have been reported. Baseline urinalysis is recommended, and patients should be monitored for the development or worsening of proteinuria. Discontinue sunitinib in patients with nephrotic syndrome.

Fistula

If fistula formation occurs, sunitinib treatment should be interrupted. Limited information is available on the continued use of sunitinib in patients with fistulae (see section 4.8).

Impaired wound healing

Cases of impaired wound healing have been reported during sunitinib therapy.

No formal clinical studies of the effect of sunitinib on wound healing have been conducted. Temporary interruption of sunitinib therapy is recommended for precautionary reasons in patients undergoing major surgical procedures. There is limited clinical experience regarding the timing of reinitiation of therapy following major surgical intervention. Therefore, the decision to resume sunitinib therapy following a major surgical intervention should be based upon clinical judgment of recovery from surgery.

Osteonecrosis of the jaw (ONJ)

Cases of ONJ have been reported in patients treated with sunitinib. The majority of cases were reported in patients who had received prior or concomitant treatment with intravenous bisphosphonates, for which ONJ is an identified risk. Caution should therefore be exercised when sunitinib and intravenous bisphosphonates are used either simultaneously or sequentially.

Invasive dental procedures are also an identified risk factor. Prior to treatment with sunitinib, a dental examination and appropriate preventive dentistry should be considered. In patients who have previously received or are receiving intravenous bisphosphonates, invasive dental procedures should be avoided if possible (see section 4.8).

Hypersensitivity/angioedema

If angioedema due to hypersensitivity occurs, sunitinib treatment should be interrupted and standard medical care provided (see section 4.8).

Seizures

In clinical studies of sunitinib and from postmarketing surveillance, seizures have been reported. Patients with seizures and signs/symptoms consistent with posterior reversible leukoencephalopathy syndrome (RPLS), such as hypertension, headache, decreased alertness, altered mental functioning and visual loss, including cortical blindness, should be controlled with medical management including control of hypertension. Temporary suspension of sunitinib is recommended; following resolution, treatment may be resumed at the discretion of the treating physician (see section 4.8).

Tumour lysis syndrome (TLS)

Cases of TLS, some fatal, have been rarely observed in clinical trials and have been reported in postmarketing surveillance in patients treated with sunitinib. Risk factors for TLS include high tumour burden, pre-existing chronic renal insufficiency, oliguria, dehydration,

hypotension, and acidic urine. These patients should be monitored closely and treated as clinically indicated, and prophylactic hydration should be considered.

Infections

Serious infections, with or without neutropenia, including some with a fatal outcome, have been reported. Uncommon cases of necrotising fasciitis, including of the perineum, sometimes fatal, have been reported (see section 4.8).

Sunitinib therapy should be discontinued in patients who develop necrotising fasciitis, and appropriate treatment should be promptly initiated.

Hypoglycaemia

Decreases in blood glucose, in some cases clinically symptomatic and requiring hospitalisation due to loss of consciousness, have been reported during sunitinib treatment. In case of symptomatic hypoglycaemia, sunitinib should be temporarily interrupted. Blood glucose levels in diabetic patients should be checked regularly in order to assess if antidiabetic medicinal product's dosage needs to be adjusted to minimise the risk of hypoglycaemia (see section 4.8).

Hyperammonaemic encephalopathy

Hyperammonaemic encephalopathy has been observed with sunitinib (see section 4.8). In patients who develop unexplained lethargy or changes in mental status, ammonia level should be measured and appropriate clinical management should be initiated.

Excipients Sodium

This medicine contains less than 1 mmol sodium (23 mg) per capsule, that is to say essentially 'sodium-free'.

4.5 Interaction with other medicinal products and other forms of interaction

Interaction studies have only been performed in adults.

Medicinal products that may increase sunitinib plasma concentrations

Effect of CYP3A4 inhibitors

In healthy volunteers, concomitant administration of a single dose of sunitinib with the potent CYP3A4 inhibitor ketoconazole resulted in an increase of the combined [sunitinib + primary metabolite] maximum concentration (C_{max}) and area under the curve ($AUC_{0-\infty}$) values of 49% and 51%, respectively.

Administration of sunitinib with potent CYP3A4 inhibitors (e.g., ritonavir, itraconazole, erythromycin, clarithromycin, grapefruit juice) may increase sunitinib concentrations.

Combination with CYP3A4 inhibitors should therefore be avoided, or the selection of an alternate concomitant medicinal product with no or minimal potential to inhibit CYP3A4 should be considered.

If this is not possible, the dose of sunitinib may need to be reduced to a minimum of 37.5 mg daily for GIST and MRCC or 25 mg daily for pNET, based on careful monitoring of tolerability (see section 4.2).

Effect of Breast Cancer Resistance Protein (BCRP) inhibitors

Limited clinical data are available on the interaction between sunitinib and BCRP inhibitors and the possibility of an interaction between sunitinib and other BCRP inhibitors cannot be excluded (see section 5.2).

Medicinal products that may decrease sunitinib plasma concentrations

Effect of CYP3A4 inducers

In healthy volunteers, concomitant administration of a single dose of sunitinib with the CYP3A4 inducer rifampicin resulted in a reduction of the combined [sunitinib + primary metabolite] $C_{\max}$ and $AUC_{0-\infty}$ values of 23% and 46%, respectively.

Administration of sunitinib with potent CYP3A4 inducers (e.g., dexamethasone, phenytoin, carbamazepine, rifampicin, phenobarbital or herbal preparations containing St. John's Wort/*Hypericum perforatum*) may decrease sunitinib concentrations. Combination with CYP3A4 inducers should therefore be avoided, or selection of an alternate concomitant medicinal product, with no or minimal potential to induce CYP3A4 should be considered. If this is not possible, the dose of sunitinib may need to be increased in 12.5 mg increments (up to 87.5 mg per day for GIST and MRCC or 62.5 mg per day for pNET), based on careful monitoring of tolerability (see section 4.2).

4.6 Fertility, pregnancy and lactation

Contraception in males and females

Women of childbearing potential should be advised to use effective contraception and avoid becoming pregnant while receiving treatment with sunitinib.

Pregnancy

There are no studies in pregnant women using sunitinib. Studies in animals have shown reproductive toxicity including foetal malformations (see section 5.3). Sunitinib should not be used during pregnancy or in women not using effective contraception, unless the potential benefit justifies the potential risk to the foetus. If sunitinib is used during pregnancy or if the patient becomes pregnant while on treatment with sunitinib, the patient should be apprised of the potential hazard to the foetus.

Breast-feeding

Sunitinib and/or its metabolites are excreted in rat milk. It is not known whether sunitinib or its primary active metabolite is excreted in human milk. Because active substances are commonly excreted in human milk and because of the potential for serious adverse reactions in breast-feeding infants, women should not breast-feed while taking sunitinib.

Fertility

Based on nonclinical findings, male and female fertility may be compromised by treatment with sunitinib (see section 5.3).

4.7 Effects on ability to drive and use machines

Sunitinib has minor influence on the ability to drive and use machines. Patients should be advised that they may experience dizziness during treatment with sunitinib.

4.8 Undesirable effects

Summary of the safety profile

The most serious adverse reactions associated with sunitinib, some fatal, are renal failure, heart failure, pulmonary embolism, gastrointestinal perforation, and haemorrhages (e.g., respiratory tract, gastrointestinal, tumour, urinary tract, and brain haemorrhages). The most common adverse reactions of any grade (experienced by patients in RCC, GIST, and pNET registrational trials) included decreased appetite, taste disturbance, hypertension, fatigue, gastrointestinal disorders (i.e. diarrhoea, nausea, stomatitis, dyspepsia, and vomiting), skin discolouration, and palmar-plantar erythrodysesthesia syndrome. These symptoms may diminish as treatment continues. Hypothyroidism may develop during treatment. Haematological disorders (e.g., neutropenia, thrombocytopenia, and anaemia) are amongst the most common adverse drug reactions.

Fatal events other than those listed in section 4.4 above or in section 4.8 below that were considered possibly related to sunitinib included multi-system organ failure, disseminated intravascular coagulation, peritoneal haemorrhage, adrenal insufficiency, pneumothorax, shock, and sudden death.

Tabulated list of adverse reactions

Adverse reactions that were reported in GIST, MRCC, and pNET patients in a pooled dataset of 7,115 patients are listed below, by system organ class, frequency and grade of severity (NCI-CTCAE). Post-marketing adverse reactions identified in clinical studies are also included. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Frequencies are defined as: very common ($\geq 1/10$), common ($(\geq 1/100$ to $< 1/10)$), uncommon ($(\geq 1/1,000$ to $< 1/100)$), rare ($(\geq 1/10,000$ to $< 1/1,000)$), very rare ($< 1/10,000$), not known (cannot be estimated from the available data).

Table 1: Adverse reactions reported in clinical trials

System organ class	Very common	Common	Uncommon	Rare	Not known
Infections and infestations	Viral infections ^a Respiratory infections ^{b,*} Abscess ^{c,*} Fungal infections ^d Urinary tract infection Skin infections ^e Sepsis ^{f,*}	Necrotising fasciitis [*] Bacterial infections ^g			
Blood and lymphatic system disorders	Neutropenia Thrombocytopenia Anaemia	Lymphopenia	Pancytopenia	Thrombotic microangiopathy ^{h,**}	

	Leukopenia				
Immune system disorders	Hypersensitivity	Angioedema			
Endocrine disorders	Hypothyroidism	Hyperthyroidism	Thyroiditis		
Metabolism and nutrition disorders	Decreased appetite ⁱ	Dehydration Hypoglycaemia	Tumour lysis syndrome*		
Psychiatric disorders	Insomnia	Depression			
Nervous system disorders	Dizziness Headache Taste disturbance ⁱ	Neuropathy peripheral Paraesthesia Hypoaesthesia Hyperaesthesia	Cerebral haemorrhage* Cerebrovascular accident* Transient ischaemic attack	Posterior reversible encephalopathy syndrome*	
Eye disorders	Periorbital oedema Eyelid oedema Lacrimation increased				
Cardiac disorders	Myocardial ischaemia ^{k,*} Ejection fraction decreased ^l	Cardiac failure congestive Myocardial infarction ^{m,*} Cardiac failure* Cardiomyopathy* Pericardial effusion Electrocardiogram QT prolonged	Left ventricular failure* Torsade de pointes		
Vascular disorders	Hypertension	Deep vein thrombosis Hot flush Flushing			
Respiratory, thoracic and mediastinal	Dyspnoea Epistaxis Cough	Pulmonary embolism* Pleural	Pulmonary haemorrhage* Respiratory		Aneurysms and artery dissections

disorders		effusion* Haemoptysis Dyspnoea exertional Oropharyngeal pain ⁿ Nasal congestion Nasal dryness	failure		*
Gastrointestinal disorders	Stomatitis o Abdominal pain ^p Vomiting Diarrhoea Dyspepsia Nausea Constipation	Gastro-oesophageal reflux disease Dysphagia Gastrointestinal haemorrhage *Oesophagitis* Abdominal distension Abdominal discomfort Rectal haemorrhage Gingival bleeding Mouth ulceration Proctalgia Cheilitis Haemorrhoids Glossodynia Oral pain Dry mouth Flatulence Oral discomfort Eructation	Gastrointestinal perforation ^{q,*} Pancreatitis Anal fistula Collitis ^r		
Hepatobiliary disorders	Hepatic failure* Cholecystitis ^s , * Hepatic function abnormal	Hepatitis			
Skin and subcutaneous	Skin discolouration ^t	Skin exfoliation	Erythema multiforme*		

tissue disorders	Palmar-plantar erythrodysesthesia syndrome Rash ^u Hair colour changes Dry skin	Skin reaction ^v Eczema Blister Erythema Alopecia Acne Pruritus Skin hyperpigmentation Skin lesion Hyperkeratosis Dermatitis Nail disorders	Stevens-Johnson syndrome* Pyoderma gangrenosum Toxic epidermal necrolysis*		
Musculoskeletal and connective tissue disorders	Pain in extremity Arthralgia Back pain	Musculoskeletal pain Muscle spasms Myalgia Muscular weakness	Osteonecrosis of Jaw Fistula*	Rhabdomyolysis* Myopathy	
Renal and urinary disorders	Renal failure* Renal failure acute* Chromaturia Proteinuria	Haemorrhage urinary tract	Nephrotic syndrome		
General disorders and administration site conditions	Mucosal inflammation Fatigue ^x Oedema ^y Pyrexia	Chest pain Pain Influenza like illness Chills	Impaired healing		
Investigations	Weight decreased White blood cell count decreased Lipase increased Platelet count decreased Haemoglobin decreased Amylase increased ^z	Blood creatine phosphokinase increased Blood thyroid stimulating hormone increased			

	Aspartate aminotransferase increased				
	Alanine aminotransferase increased				
	Blood creatinine increased				
	Blood pressure increased				
	Blood uric acid increased				

* Including fatal events.

The following terms have been combined:

a Nasopharyngitis and oral herpes.

b Bronchitis, lower respiratory tract infection, pneumonia, and respiratory tract infection.

c Abscess, abscess limb, anal abscess, gingival abscess, liver abscess, pancreatic abscess, perineal abscess, perirectal abscess, rectal abscess, subcutaneous abscess, and tooth abscess.

d Oesophageal candidiasis and oral candidiasis.

e Cellulitis and skin infection.

f Sepsis and sepsis shock.

g Abdominal abscess, abdominal sepsis, diverticulitis, and osteomyelitis.

h Thrombotic microangiopathy, thrombotic thrombocytopenic purpura, and haemolytic uraemic syndrome.

i Decreased appetite and anorexia.

j Dysgeusia, ageusia, and taste disturbance.

k Acute coronary syndrome, angina pectoris, angina unstable, coronary artery occlusion, and myocardial ischaemia.

l Ejection fraction decreased/abnormal.

m Acute myocardial infarction, myocardial infarction, and silent myocardial infarction.

n Oropharyngeal and pharyngolaryngeal pain.

o Stomatitis and aphthous stomatitis.

p Abdominal pain, abdominal pain lower, and abdominal pain upper.

q Gastrointestinal perforation and intestinal perforation.

r Colitis and colitis ischaemic.

s Cholecystitis and acalculous cholecystitis.

t Yellow skin, skin discolouration, and pigmentation disorder.

u Dermatitis psoriasiform, exfoliative rash, rash, rash erythematous, rash follicular, rash generalised, rash macular, rash maculo-papular, rash popular, and rash pruritic.

v Skin reaction and skin disorder.

w Nail disorder and discolouration.

x Fatigue and asthenic.

y Face oedema, oedema, and oedema peripheral.

z Amylase and amylase increased.

Description of selected adverse reactions

Infections and infestations

Cases of serious infection (with or without neutropenia), including cases with fatal outcome, have been reported. Cases of necrotising fasciitis, including of the perineum, sometimes fatal, have been reported (see also section 4.4).

Blood and lymphatic system disorders

Decreased absolute neutrophil counts of Grade 3 and 4 severities, respectively, were reported in 10% and 1.7% of patients on the Phase 3 GIST study, in 16% and 1.6% of patients on the Phase 3 MRCC study, and in 13% and 2.4% of patients on the Phase 3 pNET study. Decreased platelet counts of Grade 3 and 4 severities, respectively, were reported in 3.7% and 0.4% of patients on the Phase 3 GIST study, in 8.2% and 1.1% of patients on the Phase 3 MRCC study, and in 3.7% and 1.2% of patients on the Phase 3 pNET study (see section 4.4).

Bleeding events were reported in 18% of patients receiving sunitinib in a Phase 3 GIST study vs 17% of patients receiving placebo. In patients receiving sunitinib for treatment-naïve MRCC, 39% had bleeding events vs 11% of patients receiving interferon- α (IFN- α). Seventeen (4.5%) patients on sunitinib versus 5 (1.7%) patients on IFN- α experienced Grade 3 or greater bleeding events. Of patients receiving sunitinib for cytokine-refractory MRCC, 26% experienced bleeding. Bleeding events, excluding epistaxis, were reported in 21.7% of patients receiving sunitinib in the Phase 3 pNET study compared to 9.85% of patients receiving placebo (see section 4.4)

In clinical trials, tumour haemorrhage was reported in approximately 2% of patients with GIST.

Immune system disorders

Hypersensitivity reactions, including angioedema, have been reported (see section 4.4).

Endocrine disorders

Hypothyroidism was reported as an adverse reaction in 7 patients (4%) receiving sunitinib across the 2 cytokine-refractory MRCC studies; in 61 patients (16%) on sunitinib and 3 patients (< 1%) in the IFN- α arm in the treatment-naïve MRCC study.

Additionally, thyroid-stimulating hormone (TSH) elevations were reported in 4 cytokine-refractory MRCC patients (2%). Overall, 7% of the MRCC population had either clinical or laboratory evidence of treatment-emergent hypothyroidism. Acquired hypothyroidism was noted in 6.2% of GIST patients on sunitinib versus 1% on placebo. In the Phase 3 pNET study hypothyroidism was reported in 6 patients (7.2%) receiving sunitinib and in 1 patient (1.2%) on placebo.

Thyroid function was monitored prospectively in 2 studies in patients with breast cancer; sunitinib is not approved for use in breast cancer. In 1 study, hypothyroidism was reported in 15 (13.6%) patients on sunitinib and 3 (2.9%) patients on standard of care. Blood TSH increase was reported in 1 (0.9%) patient on sunitinib and no patients on standard of care. Hyperthyroidism was reported in no sunitinib-treated patients and 1 (1.0%) patient receiving standard of care. In the other study hypothyroidism was reported in a total of 31 (13%) patients on sunitinib and 2 (0.8%) patients on capecitabine. Blood TSH increase was reported in 12 (5.0%) patients on sunitinib and no patients on capecitabine. Hyperthyroidism was reported in 4 (1.7%) patients on sunitinib and no patients on capecitabine. Blood TSH decrease was reported in 3 (1.3%) patients on sunitinib and no patients on capecitabine. T4 increase was

reported in 2 (0.8%) patients on sunitinib and 1 (0.4%) patient on capecitabine. T3 increase was reported in 1 (0.8%) patient on sunitinib and no patients on capecitabine. All thyroid-related events reported were Grade 1-2 (see section 4.4).

Metabolism and nutrition disorders

A higher incidence rate of hypoglycaemia events was reported in patients with pNET in comparison to MRCC and GIST. Nevertheless, most of these adverse events observed in clinical studies were not considered related to study treatment (see section 4.4).

Nervous system disorders

In clinical studies of sunitinib and from postmarketing surveillance, there have been few reports (< 1%), some fatal, of subjects presenting with seizures and radiological evidence of RPLS. Seizures have been observed in patients with or without radiological evidence of brain metastases (see section 4.4).

Cardiac disorders

In clinical trials, decreases in left ventricular ejection fraction (LVEF) of $\geq 20\%$ and below the lower limit of normal were reported in approximately 2% of sunitinib-treated GIST patients, 4% of cytokine-refractory MRCC patients, and 2% of placebo-treated GIST patients. These LVEF declines do not appear to have been progressive and often improved as treatment continued. In the treatment-naïve MRCC study, 27% of patients on sunitinib and 15% of patients on IFN- α had an LVEF value below the lower limit of normal. Two patients (< 1%) who received sunitinib were diagnosed with CHF.

In GIST patients ‘cardiac failure’, ‘cardiac failure congestive’, or ‘left ventricular failure’ were reported in 1.2% of patients treated with sunitinib and 1% of patients treated with placebo. In the pivotal Phase 3 GIST study (N = 312), treatment-related fatal cardiac reactions were reported in 1% of patients on each arm of the study (i.e. sunitinib and placebo arms). In a Phase 2 study in cytokine-refractory MRCC patients, 0.9% of patients experienced treatment-related fatal myocardial infarction and in the Phase 3 study in treatment-naïve MRCC patients, 0.6% of patients on the IFN- α arm and 0% of patients on the sunitinib arm experienced fatal cardiac events. In the Phase 3 pNET study, 1 (1%) patient who received sunitinib had treatment-related fatal cardiac failure.

Vascular disorders

Hypertension

Hypertension was a very common adverse reaction reported in clinical trials. The dose of sunitinib was reduced or its administration temporarily suspended in approximately 2.7% of the patients who experienced hypertension. Sunitinib was not permanently discontinued in any of these patients. Severe hypertension (> 200 mmHg systolic or 110 mmHg diastolic) was reported in 4.7% of patients with solid tumours. Hypertension was reported in approximately 33.9% of patients receiving sunitinib for treatment-naïve MRCC compared to 3.6% of patients receiving IFN- α . Severe hypertension was reported in 12% of treatment-naïve patients on sunitinib and < 1% of patients on IFN- α . Hypertension was reported in 26.5% of patients receiving sunitinib in a Phase 3 pNET study, compared to 4.9% of patients receiving placebo. Severe hypertension was reported in 10% of pNET patients on sunitinib and 3% of patients on placebo.

Venous thromboembolic events

Treatment-related venous thromboembolic events were reported in approximately 1.0% of patients with solid tumours who received sunitinib on clinical trials, including GIST and RCC.

Seven patients (3%) on sunitinib and none on placebo in a Phase 3 GIST study experienced venous thromboembolic events; 5 of the 7 were Grade 3 deep venous thrombosis (DVT) and 2 were Grade 1 or 2. Four of these 7 GIST patients discontinued treatment following first observation of DVT.

Thirteen patients (3%) receiving sunitinib in the Phase 3 treatment-naïve MRCC study and 4 patients (2%) on the 2 cytokine-refractory MRCC studies had venous thromboembolic events reported. Nine of these patients had pulmonary embolisms; 1 was Grade 2 and 8 were Grade 4. Eight of these patients had DVT; 1 with Grade 1, 2 with Grade 2, 4 with Grade 3, and 1 with Grade 4. One patient with pulmonary embolism in the cytokine-refractory MRCC study experienced dose interruption.

In treatment-naïve MRCC patients receiving IFN- α , 6 (2%) venous thromboembolic events were reported; 1 patient (< 1%) experienced a Grade 3 DVT and 5 patients (1%) had pulmonary embolisms, all with Grade 4.

Venous thromboembolic events were reported for 1 (1.2%) patient in the sunitinib arm and 5 (6.1%) patients in the placebo arm in the Phase 3 pNET study. Two of these patients on placebo had DVT, 1 with Grade 2 and 1 with Grade 3.

No cases with fatal outcome were reported in GIST, MRCC, and pNET registrational studies. Cases with fatal outcome have been observed in the postmarketing surveillance.

Cases of pulmonary embolism were observed in approximately 3.1% of patients with GIST and in approximately 1.2% of patients with MRCC, who received sunitinib in Phase 3 studies. No pulmonary embolism was reported for patients with pNET who received sunitinib in the Phase 3 study. Rare cases with fatal outcome have been observed in the postmarketing surveillance.

Patients who presented with pulmonary embolism within the previous 12 months were excluded from sunitinib clinical studies.

In patients who received sunitinib in Phase 3 registrational studies, pulmonary events (i.e. dyspnoea, pleural effusion, pulmonary embolism, or pulmonary oedema) were reported in approximately 17.8% of patients with GIST, in approximately 26.7% of patients with MRCC and in 12% of patients with pNET.

Approximately 22.2% of patients with solid tumours, including GIST and MRCC, who received sunitinib in clinical trials experienced pulmonary events.

Gastrointestinal disorders

Pancreatitis has been observed uncommonly (< 1%) in patients receiving sunitinib for GIST or MRCC. No treatment-related pancreatitis was reported in the Phase 3 pNET study (see section 4.4).

Fatal gastrointestinal bleeding was reported in 0.98% of patients receiving placebo in the GIST Phase 3 study.

Hepatobiliary disorders

Hepatic dysfunction has been reported and may include Liver Function Test abnormalities, hepatitis, or liver failure (see section 4.4).

Skin and subcutaneous tissue disorders

Cases of pyoderma gangrenosum, generally reversible after discontinuation of sunitinib, have been reported (see also section 4.4).

Musculoskeletal and connective tissue disorders

Cases of myopathy and/or rhabdomyolysis, some with acute renal failure, have been reported. Patients with signs or symptoms of muscle toxicity should be managed as per standard medical practice (see section 4.4).

Cases of fistula formation, sometimes associated with tumour necrosis and regression, in some cases with fatal outcomes, have been reported (see section 4.4).

Cases of ONJ have been reported in patients treated with sunitinib, most of which occurred in patients who had identified risk factors for ONJ, in particular, exposure to intravenous bisphosphonates and/or a history of dental disease requiring invasive dental procedures (see also section 4.4).

Investigations

Data from non-clinical (*in vitro* and *in vivo*) studies, at doses higher than the recommended human dose, indicated that sunitinib has the potential to inhibit the cardiac action potential repolarisation process (e.g., prolongation of QT interval).

Increases in the QTc interval to over 500 msec were reported in 0.5%, and changes from baseline in excess of 60 msec were reported in 1.1% of the 450 solid tumour patients; both of these parameters are recognised as potentially significant changes. At approximately twice therapeutic concentrations, sunitinib has been shown to prolong the QTcF interval (Fridericia corrected QT interval).

QTc interval prolongation was investigated in a trial in 24 patients, ages 20-87 years, with advanced malignancies. The results of this study demonstrated that sunitinib had an effect on QTc interval (defined as a mean placebo-adjusted change of > 10 msec with a 90% confidence interval [CI] upper limit > 15 msec) at therapeutic concentration (Day 3) using the within-day baseline correction method, and at greater than therapeutic concentration (Day 9) using both baseline correction methods. No patients had a QTc interval > 500 msec. Although an effect on QTcF interval was observed on Day 3 at 24 hours postdose (i.e., at therapeutic plasma concentration expected after the recommended starting dose of 50 mg) with the within-day baseline correction method, the clinical significance of this finding is unclear.

Using comprehensive serial ECG assessments at times corresponding to either therapeutic or greater than therapeutic exposures, none of the patients in the evaluable or intent-to-treat (ITT) populations were observed to develop QTc interval prolongation considered as “severe”

(i.e. equal to or greater than Grade 3 by Common Terminology Criteria for Adverse Events [CTCAE] version 3.0).

At therapeutic plasma concentrations, the maximum QTcF interval (Frederica's correction) mean change from baseline was 9 msec (90% CI: 15.1 msec). At approximately twice therapeutic concentrations, the maximum QTcF interval change from baseline was 15.4 msec (90% CI: 22.4 msec). Moxifloxacin (400 mg) used as a positive control showed a 5.6 msec maximum mean QTcF interval change from baseline. No subjects experienced an effect on the QTc interval greater than Grade 2 (CTCAE version 3.0) (see section 4.4).

Long-term safety in MRCC

The long-term safety of sunitinib in patients with MRCC was analysed across 9 completed clinical studies conducted in the first-line, bevacizumab-refractory, and cytokine-refractory treatment settings in 5,739 patients, of whom 807 (14%) were treated for ≥ 2 years up to 6 years. In the 807 patients who received long-term sunitinib treatment, most treatment-related adverse events (TRAEs) occurred initially in the first 6 months–1 year and then were stable or decreased in frequency over time, with the exception of hypothyroidism, which gradually increased over time, with new cases occurring over the 6 year period. Prolonged treatment with sunitinib did not appear to be associated with new types of TRAEs.

Paediatric population

The safety profile of sunitinib has been derived from a Phase 1 dose-escalation study, a Phase 2 open-label study, a Phase 1/2 single-arm study and from publications as described below.

A Phase 1 dose-escalation study of oral sunitinib was conducted in 35 patients comprised of 30 paediatric patients (aged 3 years to 17 years) and 5 young adult patients (aged 18 to 21 years), with refractory solid tumours, the majority of whom had a primary diagnosis of brain tumour. All study participants experienced adverse drug reactions; most of these were severe (toxicity grade ≥ 3) and included cardiac toxicity. The most common adverse drug reactions were gastrointestinal (GI) toxicity, neutropenia, fatigue, and ALT elevation. The risk of cardiac adverse drug reactions appeared to be higher in paediatric patients with previous exposure to cardiac irradiation or anthracycline compared to those paediatric patients without previous exposure. In these paediatric patients without previous exposure to anthracyclines or cardiac irradiation, the maximum tolerated dose (MTD) has been identified (see section 5.1).

A phase 2 open-label study was conducted in 29 patients comprised of 27 paediatric patients (aged 3 years to 16 years) and 2 young adult patients (aged 18 years to 19 years) with recurrent/progressive/refractory high grade glioma (HGG) or ependymoma. There were no Grade 5 adverse reactions in either group. The most common ($\geq 10\%$) treatment-related adverse events were neutrophil count decreased (6 [20.7%] patients) and haemorrhage intracranial (3[10.3%] patients).

A Phase 1/2 single-arm, study was conducted in 6 paediatric patients (aged 13 years to 16 years) with advanced unresectable GIST. The most frequent adverse drug reactions were diarrhoea, nausea, WBC count decreased, neutropenia, and headache in 3 (50.0%) patients each, primarily Grade 1 or 2 in severity. Four out of 6 patients (66.7%) experienced Grade 3-4 treatment-related adverse events (Grade 3 hypophosphataemia, neutropenia, and thrombocytopenia in 1 patient each and a Grade 4 neutropenia in 1 patient). There were no serious adverse events (SAEs) or Grade 5 adverse drug reactions reported in this study. In

both the clinical study and the publications, the safety profile was consistent with the known safety profile in adults.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the Yellow Card Scheme at: www.mhra.gov.uk/yellowcard or search for MHRA Yellow Card in the Google Play or Apple App Store.

4.9 Overdose

There is no specific antidote for overdose with sunitinib and treatment of overdose should consist of general supportive measures. If indicated, elimination of unabsorbed active substance may be achieved by emesis or gastric lavage. Cases of overdose have been reported; some cases were associated with adverse reactions consistent with the known safety profile of sunitinib.

5 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antineoplastic agent protein kinase inhibitor, ATC code: L01XE01

Mechanism of action

Sunitinib inhibits multiple RTKs that are implicated in tumour growth, neoangiogenesis, and metastatic progression of cancer. Sunitinib was identified as an inhibitor of platelet-derived growth factor receptors (PDGFR α and PDGFR β), VEGF receptors (VEGFR1, VEGFR2, and VEGFR3), stem cell factor receptor (KIT), Fms-like tyrosine kinase-3 (FLT3), colony stimulating factor receptor (CSF-1R), and the glial cell-line derived neurotrophic factor receptor (RET). The primary metabolite exhibits similar potency compared to sunitinib in biochemical and cellular assays.

Sunitinib inhibited the phosphorylation of multiple RTKs (PDGFR β , VEGFR2, KIT) in tumor xenografts expressing RTK targets in vivo and demonstrated inhibition of tumor growth or tumor regression, and/or inhibited in metastases in some experimental models of cancer. Sunitinib demonstrated the ability to inhibit growth of tumor cells expressing dysregulated target RTKs (PDGFR, RET, or KIT) in vitro and to inhibit PDGFR β - and VEGFR2-dependent tumor angiogenesis in vivo.

The clinical safety and efficacy of sunitinib has been studied in subjects with malignant GIST who were resistant to imatinib (i.e., those who experienced disease progression during or following treatment with imatinib); or intolerant to imatinib (i.e., those who experienced significant toxicity during treatment with imatinib that precluded further treatment); in subjects with advanced renal cell carcinoma (RCC); and in subjects with unresectable pNET.

Efficacy is based on time to tumor progression and an increase in survival in GIST.

Efficacy is based on progression-free survival (PFS) and objective response rates.

5.2 Pharmacokinetic properties

The PK of sunitinib was evaluated in 135 healthy volunteers and 266 patients with solid tumours. The PK was similar in all solid tumours populations tested and in healthy volunteers.

In the dosing ranges of 25 to 100 mg, the area under the plasma concentration-time curve (AUC) and $C_{\max}$ increased proportionally with dose. With repeated daily administration, sunitinib accumulates 3- to 4-fold and its primary active metabolite accumulates 7- to 10-fold. Steady-state concentrations of sunitinib and its primary active metabolite are achieved within 10 to 14 days. By Day 14, combined plasma concentrations of sunitinib and its active metabolite are 62.9-101 ng/ml, which are target concentrations predicted from preclinical data to inhibit receptor phosphorylation *in vitro* and result in tumour stasis/growth reduction *in vivo*. The primary active metabolite comprises 23% to 37% of the total exposure. No significant changes in the PK of sunitinib or the primary active metabolite are observed with repeated daily administration or with repeated cycles in the dosing schedules tested.

Absorption

After oral administration of sunitinib, $C_{\max}$ are generally observed from 6 to 12 hours time to maximum concentration ($t_{\max}$) post-administration.

Food has no effect on the bioavailability of sunitinib.

Distribution

In vitro, binding of sunitinib and its primary active metabolite to human plasma protein was 95% and 90%, respectively, with no apparent concentration dependence. The apparent volume of distribution (V_d) for sunitinib was large, 2230 L, indicating distribution into the tissues.

Metabolic interactions

The calculated *in vitro* K_i values for all cytochrome P450 (CYP) isoforms tested (CYP1A2, CYP2A6, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6, CYP2E1, CYP3A4/5, and CYP4A9/11) indicated that sunitinib and its primary active metabolite are unlikely to induce metabolism, to any clinically relevant extent, of other active substances that may be metabolised by these enzymes.

Biotransformation

Sunitinib is metabolised primarily by CYP3A4, the CYP isoform which produces its primary active metabolite, desethyl sunitinib, which is then further metabolised by the same isoenzyme.

Co-administration of sunitinib with potent CYP3A4 inducers or inhibitors should be avoided because the plasma levels of sunitinib may be altered (see sections 4.4 and 4.5).

Elimination

Excretion is primarily via faeces (61%), with renal elimination of unchanged active substance and metabolites accounting for 16% of the administered dose. Sunitinib and its primary active metabolite were the major compounds identified in plasma, urine, and faeces, representing 91.5%, 86.4%, and 73.8% of radioactivity in pooled samples, respectively. Minor metabolites were identified in urine and faeces, but generally were not found in plasma. Total oral

clearance (CL/F) was 34-62 L/h. Following oral administration in healthy volunteers, the elimination half-lives of sunitinib and its primary active desethyl metabolite are approximately 40-60 hours and 80-110 hours, respectively.

Co-administration with medicinal products that are BCRP inhibitors

In vitro, sunitinib is a substrate of the efflux transporter BCRP. In study A6181038 the co-administration of gefitinib, a BCRP inhibitor, did not result in a clinically relevant effect on the C_{max} and AUC for sunitinib or total drug (sunitinib + metabolite) (see section 4.5). This study was a multi-centre, open-label, Phase 1/2 study examining the safety/tolerability, the maximum tolerated dose, and the antitumour activity of sunitinib in combination with gefitinib in subjects with MRCC. The PK of gefitinib (250 mg daily) and sunitinib (37.5 mg [Cohort 1, n=4] or 50 mg [Cohort 2, n=7] daily on a 4-weeks on followed by 2 weeks-off schedule) when co-administered was evaluated as a secondary study objective. Changes in sunitinib PK parameters were of no clinical significance and did not indicate any drug-drug interactions; however, considering the relatively low number of subjects (i.e. N=7+4) and the moderate-large interpatient variability in the pharmacokinetic parameters, caution needs to be taken when interpreting the PK drug-drug interaction findings from this study.

Special populations

Hepatic impairment

Sunitinib and its primary metabolite are mainly metabolised by the liver. Systemic exposures after a single dose of sunitinib were similar in subjects with mild or moderate (Child-Pugh Class A and B) hepatic impairment compared to subjects with normal hepatic function. Sunitinib was not studied in subjects with severe (Child-Pugh Class C) hepatic impairment.

Studies in cancer patients have excluded patients with ALT or AST > 2.5 x ULN (upper limit of normal) or > 5.0 x ULN if due to liver metastasis.

Renal impairment

Population PK analyses indicated that sunitinib apparent clearance (CL/F) was not affected by creatinine clearance (CL_{cr}) within the range evaluated (42-347 ml/min). Systemic exposures after a single dose of sunitinib were similar in subjects with severe renal impairment (CL_{cr} < 30 ml/min) compared to subjects with normal renal function (CL_{cr} > 80 ml/min). Although sunitinib and its primary metabolite were not eliminated through haemodialysis in subjects with ESRD, the total systemic exposures were lower by 47% for sunitinib and 31% for its primary metabolite compared to subjects with normal renal function.

Weight, performance status

Population PK analyses of demographic data indicate that no starting dose adjustments are necessary for weight or Eastern Cooperative Oncology Group (ECOG) performance status.

Gender

Available data indicate that females could have about 30% lower apparent clearance (CL/F) of sunitinib than males: this difference, however, does not necessitate starting dose adjustments.

Paediatric population

Experience on the use of sunitinib in paediatric patients is limited (see section 4.2). Population PK analyses of a pooled dataset from adult patients with GIST and solid tumours and

paediatric patients with solid tumours were completed. Stepwise covariate modelling analyses were performed to evaluate the effect of age and body size (total body weight or body surface area) as well as other covariates on important PK parameters for sunitinib and its active metabolite. Among age and bodysize related covariates tested, age was a significant covariate on apparent clearance of sunitinib (the younger the age of the paediatric patient, the lower the apparent clearance). Similarly, body surface area was a significant covariate on the apparent clearance of the active metabolite (the lower the body surface area, the lower the apparent clearance).

Furthermore, based on an integrated population PK analysis of pooled data from the 3 paediatric studies (2 paediatric solid tumour studies and 1 paediatric GIST study; ages: 6 years to 11 years and 12 years to 17 years), baseline body surface area (BSA) was a significant covariate on apparent clearance of sunitinib and its active metabolite. Based on this analysis, a dose of approximately 20 mg/m² daily in paediatric patients, with BSA values between 1.10 and 1.87 m², is expected to provide plasma exposures to sunitinib and its active metabolite comparable (between 75 and 125% of the AUC) to those in adults with GIST administered sunitinib 50 mg daily on Schedule 4/2 (AUC 1233 ng.hr/mL). In paediatric studies, the starting dose of sunitinib was 15 mg/m² (based on the MTD identified in the Phase 1 dose-escalation study, see section 5.1), which in paediatric patients with GIST increased to 22.5 mg/m² and subsequently to 30 mg/m² (not to exceed the total dose of 50 mg/day) based on individual patient safety/tolerability. Furthermore, according to the published literatures in paediatric patients with GIST, the calculated starting dose ranged from 16.6 mg/m² to 36 mg/m², increased to doses as high as 40.4 mg/m² (not exceeding the total dose of 50 mg/day).

6 PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Capsule content

Mannitol (E421)

Croscarmellose sodium

Starch, pregelatinized (corn)

Magnesium stearate

Capsule shell

Gelatin Purified water

Titanium dioxide (E171)

Red iron oxide (E172)

Printing ink

Shellac (E904)

Propylene glycol (E1520)

Potassium hydroxide (E525)

Titanium dioxide (E171)

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

2 years.

6.4 Special precautions for storage

Store below 30°C.

6.5 Nature and contents of container

10 Capsules of Sunitinib Capsules are packed with plain Aluminium foil on one side and PVC/Aclar film on other side in the form of a blister pack such 3 blisters are further packed in printed carton along with instructions for use.

6.6 Special precautions for disposal

No special requirements.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7 MARKETING AUTHORISATION HOLDER

Product Registration Holder in Malaysia:

MEGA LIFESCIENCES SDN BHD

B-28-02, The Ascent, Paradigm.

No.1, Jalan SS7/26A, Kelana Jaya, 47301 Petaling Jaya, Selangor, Malaysia.

Manufactured by:

MSN LABORATORIES PRIVATE LIMITED,

Formulation Division, Unit-II, Sy.no. 1277, 1319 to 1324, Nandigama (Village & Mandal), Rangareddy (District)Telangana-509 228, India.

8 DATE OF REVISION OF THE TEXT

October 2025